What does a quantum reservoir bring to physics-informed solid mechanics?

A fair benchmark against classical random features

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Abstract

Quantum physics-informed neural networks (QPINNs) are often reported with few parameters and modest accuracy, but rarely against classical models of the same size that have been tuned with the same care. We revisit the question for linear solid mechanics. First, variational QPINNs lose to size-matched classical networks on Lamé, Kirsch and thermoelastic problems, and the bottleneck is optimization rather than expressivity. Second, we remove the optimization by freezing the circuit and solving only a linear readout by one least-squares solve (a quantum extreme-learning or reservoir PINN, QERC-PINN). With a per-qubit encoding and a Haar or Clifford+T reservoir, QERC-PINN reaches relative errors of $\sim 5 \times 10^{-5}$ on the Kirsch plate and resolves the stress concentration factor $K_t = 3.00$. Third, we show that the features $p_k(\boldsymbol{x}) = \mathrm{Tr}[U^\dagger |k\rangle\langle k| U \rho(\boldsymbol{x})]$ are linear in the encoded state, so the reservoir only selects a random 2^N-dimensional subspace of the tensor-product function space spanned by $\rho(\boldsymbol{x})$; the encoding alone fixes what can be represented. A classical random projection of the same tensor-product space reproduces the quantum results exactly. On application problems with material interfaces — a through-silicon via (TSV) and periodic TSV arrays — the quantum reservoir matches a classical two-layer random network, while cosine (Fourier) random features are best in almost every case, by up to $\sim 40\times$. Finally, measurement shot noise is the main obstacle on hardware: with 10^7 shots per circuit the strong-form solution still has errors of 4–11%. Moving to an energy (Ritz) formulation that needs only first derivatives reduces the error 3–4 times (about one order of magnitude in shots), but a further 2–3 orders of magnitude are needed. We release all code and data.

1 Introduction

Physics-informed neural networks (PINNs) [1] and their quantum counterparts (QPINNs) [**? ?**] are attractive for small-parameter surrogate models. Claims of parameter efficiency, however, depend strongly on the classical baseline. In this work we ask a narrower question with careful controls: *for linear elasticity, what part of a quantum model actually helps, and can a classical model with the same structure reproduce it?*

Our contributions are:

1. A negative but diagnostic result for variational QPINNs: the bottleneck is optimization (Sec. 3).
2. A QERC-PINN that avoids optimization by one least-squares solve, and a simple argument that the reservoir cannot change the representable function space (Sec. 4).
3. A fair benchmark with tuned classical random features, including cosine features and classical controls that share the tensor-product structure of the quantum features (Secs. 5–6).
4. A shot-noise analysis and a first mitigation by an energy formulation (Sec. 7).

Table 1: Cross-check of the reference solutions (relative L_2 error of displacement; von Mises as mean absolute error normalised by the maximum, as in Zhu et al. [4]).

Comparison	$\boldsymbol{u}$	von Mises, mid-plane	von Mises, near top
Axisymmetric, clamped: our FE vs Abaqus	2.2×10^{-4}	3.2×10^{-4}	6.5×10^{-4}
Axisymmetric, free top: our FE vs Abaqus	8.8×10^{-5}	3.2×10^{-4}	7.7×10^{-4}
3D clamped mid-plane vs 2D plane strain	2.4×10^{-4}	3.1×10^{-4}	–

2 Methods

2.1 Problems

All problems are 2D linear elasticity in plane strain or plane stress: $\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}$, $\boldsymbol{\sigma} = \lambda \operatorname{tr}\boldsymbol{\varepsilon}\, \boldsymbol{I} + 2\mu\boldsymbol{\varepsilon} - \beta\Delta T\, \boldsymbol{I}$, $\beta = (3\lambda + 2\mu)\alpha$.

- **Kirsch plate**: an infinite plate with a circular hole under uniaxial tension, truncated at R, with the exact displacement on the outer boundary and traction-free hole ($K_t = 3$).
- **Single TSV**: a Cu via (radius a) in Si (outer radius $4a$), uniform cooling $\Delta T = -250\,\mathrm{K}$.
- **Periodic TSV array**: square array with pitch L; a quarter cell with symmetric roller conditions (zero normal displacement and zero shear traction) under zero macroscopic strain.
- **TSV unit cell in 3D and in axisymmetry**: the geometry of Zhu et al. [4] (via diameter 5 µm, height 50 µm, pitch 15 µm, $\Delta T = -250\,\mathrm{K}$, liner omitted). We solve the upper half with a symmetric mid-plane and either a clamped top (their scenario 1) or a traction-free top (the wafer surface). The axisymmetric model replaces the square cell by a cylinder of equal area; the 3D model is the quarter of the square cell.

References are analytic (Kirsch, single TSV) or a P1 finite-element solution on an interface-fitted mesh with $h = 0.005a$, checked against $2h$ (differences $\leq 3 \times 10^{-5}$ in displacement). For the 3D and axisymmetric cells we computed Abaqus references (CAX8, C3D10) and cross-checked them (Table 1): our axisymmetric FE and Abaqus agree to $< 10^{-3}$, the mid-plane stress at the via centre matches the plane-strain Lamé solution exactly ($\sigma_{rr} = \sigma_{\theta\theta} = 642\,\mathrm{MPa}$, $\sigma_{zz} = 917\,\mathrm{MPa}$), and the mid-plane of the clamped 3D cell coincides with the 2D plane-strain cell.

2.2 Least-squares PINN with fixed features

The displacement in each material phase is $\boldsymbol{u}(\boldsymbol{x}) = \sum_k \boldsymbol{c}_k \phi_k(\boldsymbol{x})$ with fixed features ϕ_k. PDE residuals at collocation points, boundary conditions and interface conditions (continuity of $\boldsymbol{u}$ and of traction, with the thermal-mismatch jump) are all linear in $\boldsymbol{c}$, so the coefficients follow from one regularized least-squares solve [5]. Each phase has its own features.

2.3 Features compared

- **QERC (quantum)**: per-qubit encoding $|\psi(\boldsymbol{x})\rangle = \bigotimes_{l=1}^{N} \left(\cos\frac{\theta_l}{2}|0\rangle + e^{i\varphi_l}\sin\frac{\theta_l}{2}|1\rangle\right)$, with $2N$ angles from a random affine map of the input (as in Ikeda et al. [6]), followed by a fixed reservoir U (Haar, Clifford+T or transverse-field Ising evolution); the 2^N outcome probabilities $p_k(\boldsymbol{x}) = |\langle k|U|\psi(\boldsymbol{x})\rangle|^2$ are the features.
- **tanh RF**: $\tanh(\boldsymbol{w}_k \cdot \boldsymbol{x} + b_k)$ [7].
- **Fourier RF / Fourier PIELM**: $\cos(\boldsymbol{w}_k \cdot \boldsymbol{x} + b_k)$; FS-PIELM [**?**].

- **Tensor random projection (classical control)**: $\mathrm{Re}\,\langle\psi(\boldsymbol{x})|R_k|\psi(\boldsymbol{x})\rangle$ with the same $|\psi(\boldsymbol{x})\rangle$ and independent GUE matrices R_k.
- **MLP2 (classical control)**: features of a random two-layer tanh network with $2N$ hidden units.

Every model has the same number of features (2^N) and its input scale is swept; for each seed we report the best scale, and we widen the sweep until the best scale lies inside it. This tuning was decisive: an untuned classical baseline looked equal to the quantum model with two seeds, while the tuned one was 20× better; and at $N = 12$ the cosine features improved from 3.2×10^{-4} to 8.3×10^{-6} once the sweep was extended beyond its original upper edge.

The quantum features are simulated exactly as state vectors in JAX, which provides the first and second derivatives with respect to the coordinates by automatic differentiation. We checked the implementation against Qiskit: for $N = 2$–6 the outcome probabilities agree to 4×10^{-16} (Statevector), and with 2×10^5 shots (Aer, $N = 4$) the sampled frequencies agree within 2.1 standard errors.

3 Variational QPINNs are limited by optimization

On the 1D Lamé problem, a supervised fit of the analytic solution reaches 7×10^{-4} while PDE training of the same circuit stops at 5×10^{-2}. On the Kirsch plate, polar or Chebyshev encodings improve the displacement error by 2–3 times, but a classical MLP of the same size with the same inputs stays ahead. Since the analytic solution lies close to the circuit's function class, we attribute the gap to optimization.

4 What the reservoir can and cannot add

Write the per-qubit encoding as $\rho(\boldsymbol{x}) = \bigotimes_{l=1}^{N} \rho_l(\boldsymbol{x})$, $\rho_l = |q_l\rangle\langle q_l|$, $|q_l\rangle = \cos\frac{\theta_l}{2}|0\rangle + e^{i\theta_{N+l}}\sin\frac{\theta_l}{2}|1\rangle$, and let

$$\mathcal{F}_\rho = \left\{\boldsymbol{x} \mapsto \mathrm{Tr}[H\rho(\boldsymbol{x})] \;:\; H = H^\dagger \in \mathbb{C}^{2^N \times 2^N}\right\} \tag{1}$$

be the real function space spanned by the matrix elements of $\rho(\boldsymbol{x})$.

Proposition 1 (the reservoir selects a subspace)**.** *For any unitary U, the features $p_k(\boldsymbol{x}) = |\langle k|U|\psi(\boldsymbol{x})\rangle|^2$ satisfy* $\mathrm{span}\{p_k\} = \{\mathrm{Tr}[H\rho] : H \in \mathcal{D}_U\} \subseteq \mathcal{F}_\rho$*, where $\mathcal{D}_U = \mathrm{span}_{\mathbb{R}}\{U^\dagger|k\rangle\langle k|U\}$ is a commutative algebra of dimension 2^N. Hence* $\dim\mathrm{span}\{p_k\} \le \min(2^N, \dim\mathcal{F}_\rho)$*, and $\sum_k p_k \equiv 1$.*

Proof. $p_k = \mathrm{Tr}[M_k\rho]$ with $M_k = U^\dagger|k\rangle\langle k|U$ and the trace is linear in M_k; the M_k are orthogonal projectors summing to I. □

Proposition 2 (product structure of $\mathcal{F}_\rho$)**.** $\mathcal{F}_\rho = \mathrm{span}\big\{\prod_{l=1}^{N}\mathrm{Tr}[A_l\rho_l(\boldsymbol{x})] \;:\; A_l = A_l^\dagger \in \mathbb{C}^{2\times 2}\big\}$*, and each factor lies in* $\mathrm{span}\{1,\ \boldsymbol{n}_l(\boldsymbol{x})\}$*, where $\boldsymbol{n}_l = (\sin\theta_l\cos\theta_{N+l}, \sin\theta_l\sin\theta_{N+l}, \cos\theta_l)$ is the Bloch vector of qubit l. In particular* $\dim\mathcal{F}_\rho \le 4^N$.

Proof. Product operators $\bigotimes_l A_l$ span the Hermitian matrices over $\mathbb{R}$, and $\mathrm{Tr}[(\bigotimes_l A_l)\bigotimes_l \rho_l] = \prod_l \mathrm{Tr}[A_l\rho_l]$. Writing $A_l = a_0 I + \boldsymbol{a}\cdot\boldsymbol{\sigma}$ gives $\mathrm{Tr}[A_l\rho_l] = a_0 + \boldsymbol{a}\cdot\boldsymbol{n}_l$. □

Corollary 1 (classical access to the same space)**.** *The product random features $f_k(\boldsymbol{x}) = \prod_l \mathrm{Tr}[A_{k,l}\rho_l(\boldsymbol{x})]$ with random Hermitian $A_{k,l}$ lie in $\mathcal{F}_\rho$ and cost $O(N)$ operations each, without a 2^N-dimensional state. The tensor random projection of Sec. 6 ($\mathrm{Re}\,\langle\psi|R_k|\psi\rangle$, GUE R_k) also lies in $\mathcal{F}_\rho$ but costs $O(4^N)$ per feature.*

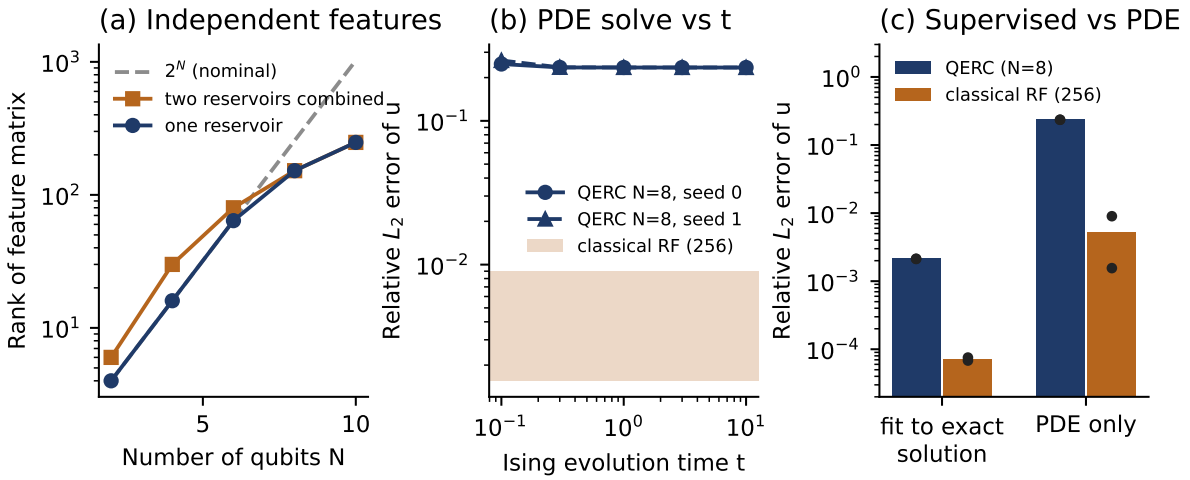


Figure 1: Effect of the reservoir for per-qubit encoding (Kirsch, PDE only). TODO: update caption to the figure.

Proposition 3 (readout cost on hardware)**.** *Let each p_k be estimated from S shots. Then* $\operatorname{Var} \hat{p}_k = p_k(1-p_k)/S$*; for a Haar reservoir* $\mathbb{E}\, p_k = 2^{-N}$*, so the relative standard deviation is* $\approx \sqrt{2^N/S}$ *and a relative accuracy ε on the features needs $S \gtrsim 2^N/\varepsilon^2$ shots per circuit. The second derivative $\partial^2 p_k/\partial x^2 = \sum_{ij} \partial_{\theta_i\theta_j} p_k\, \theta_i'\theta_j' + \sum_i \partial_{\theta_i} p_k\, \theta_i''$ requires $O(N^2)$ parameter-shifted circuits, so one collocation point costs $O(N^2\, 2^N/\varepsilon^2)$ shots.*

Proof. Multinomial sampling; the parameter-shift rule [9, 10] gives each $\partial_{\theta_i\theta_j} p_k$ from four shifted circuits, and there are $(2N)^2$ pairs. □

Corollary 2 (no advantage from feature evaluation)**.** *For $m = 2^N$ features at M points, the quantum readout costs $O(MN^2 2^N/\varepsilon^2)$ shots, while the classical product features of Corollary 1 cost $O(MN2^N)$ operations, with the same function space $\mathcal{F}_\rho$ available to both. The ratio is $N/\varepsilon^2 \geq 1$ for any useful $\varepsilon < 1$.*

Together, Propositions 1–3 say that (i) what a quantum reservoir PINN can represent is fixed by the encoding, (ii) the same space is classically accessible at linear cost in N, and (iii) reading it out on hardware is more expensive than computing it classically. A benefit of the quantum device can therefore only come from the particular 2^N-dimensional subspace that a reservoir selects; Sections 5–6 test whether that subspace is better than random ones.

Remark. Proposition 3 concerns outcome probabilities. Observables with $O(1)$ expectation values (e.g. a few Pauli strings) avoid the 2^N factor in S but give fewer features; this is left for future work.

Two numerical consequences are confirmed in Fig. 1: (i) if every qubit receives the same input, $\dim \mathcal{F}_\rho$ is small and the reservoir does not matter (changing the Ising time or couplings leaves the PDE error unchanged to three digits; at $N = 8$ only 152 of 256 features are independent); (ii) with a per-qubit encoding, $\dim \mathcal{F}_\rho$ grows like 4^N and a generic reservoir (Clifford+T, Ising, Haar) improves the error by two orders of magnitude over no reservoir or a Clifford-only reservoir, in line with Ikeda et al. [6]: a reservoir that mixes the computational basis is needed to select a rich subspace, but any generic one does.

5 Fair comparison on the Kirsch plate

With tuned scales on both sides (Table 2, Fig. 2), the quantum reservoir and tanh random features are equivalent, and both recover $K_t = 3.00 \pm 0.03$. Cosine features — the classical

Table 2: Kirsch plate, PDE only, $N = 10$ (1024 features), 5 seeds, best input scale per seed. Relative L_2 error of displacement: geometric mean [min, max].

Features	rel. $L_2(\boldsymbol{u})$
QERC, Haar	4.6×10^{-5} [2.9×10^{-5}, 8.4×10^{-5}]
QERC, Clifford+T	4.7×10^{-5} [2.4×10^{-5}, 9.7×10^{-5}]
tanh RF	5.9×10^{-5} [3.2×10^{-5}, 1.0×10^{-4}]
Fourier PIELM	$\mathbf{1.1 \times 10^{-6}}$ [4.8×10^{-7}, 1.9×10^{-6}]
FS-PIELM	1.3×10^{-6} [4.4×10^{-7}, 6.2×10^{-6}]

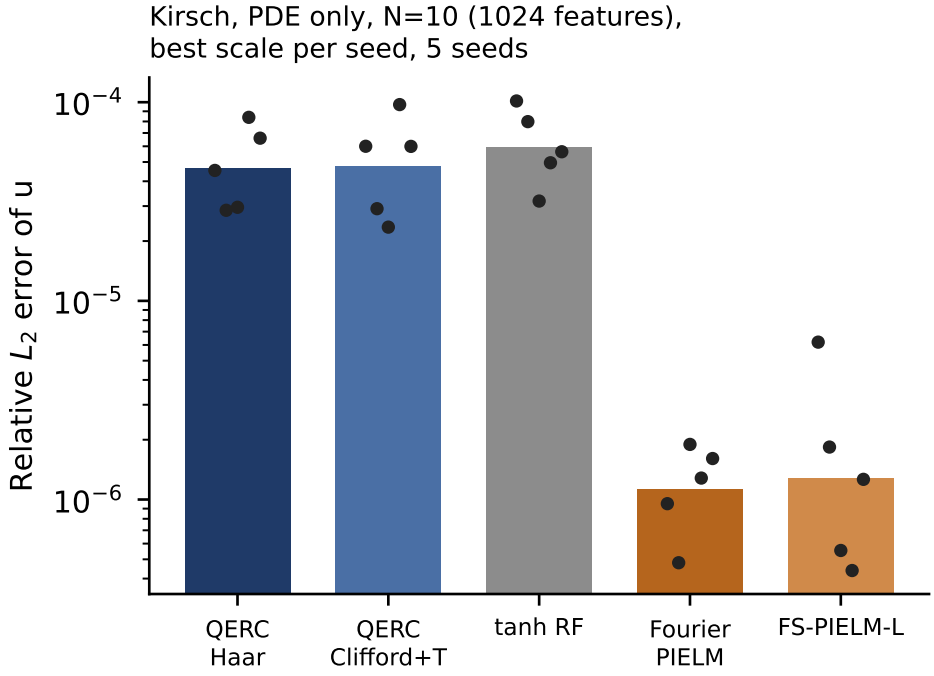


Figure 2: Kirsch plate: all feature types with tuned input scale (5 seeds).

analogue closest to the trigonometric quantum features — are about 40× more accurate.

6 Problems with material interfaces

6.1 Single TSV

For a single Cu via in Si ($N = 8$, 5 seeds), QERC reaches 4.6×10^{-3}, ten times better than one-layer tanh (4.8×10^{-2}) or Fourier (4.3×10^{-2}) features. The tensor random projection reproduces the quantum result (4.4×10^{-3}, with matching per-seed and per-scale trends; Fig. 3), and a random two-layer network reaches 1.2×10^{-2}. The advantage over one-layer features therefore comes from the product structure of $\mathcal{F}_\rho$, not from anything specific to quantum measurement. The advantage also disappears with more features: at $N = 10$ QERC gives 1.5×10^{-4} (tanh 2.1×10^{-3}, Fourier 1.5×10^{-3}), and at $N = 12$ the cosine features with a properly extended scale sweep give 8.3×10^{-6}, about 7× better than QERC (5.6×10^{-5}) and tanh (5.7×10^{-5}).

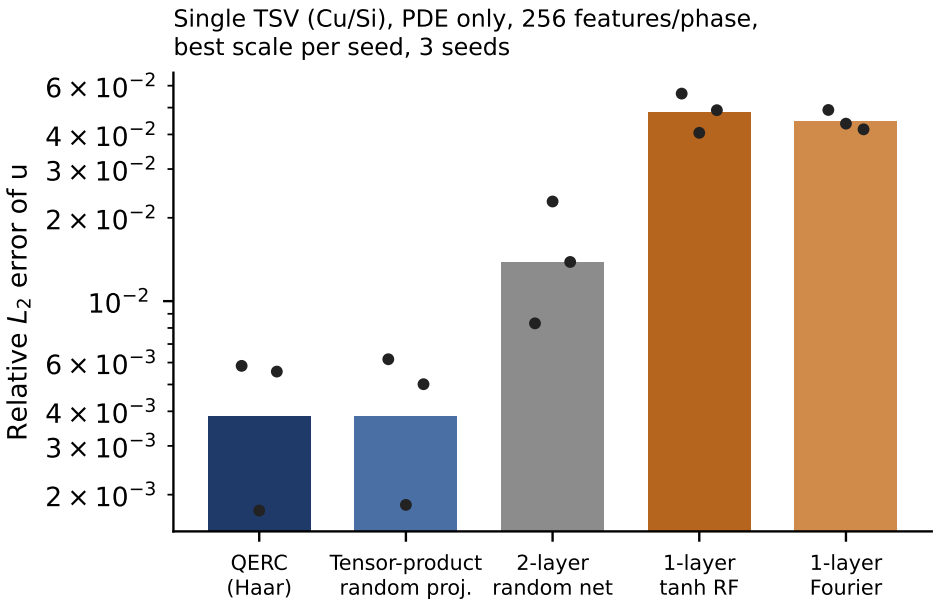


Figure 3: Single TSV: quantum reservoir and classical controls.

Table 3: Axisymmetric TSV cell, $N = 8$, 5 seeds: relative $L_2(\boldsymbol{u})$ and normalised von Mises error near the top.

	clamped top		free top	
Features	$\boldsymbol{u}$	von Mises, top	$\boldsymbol{u}$	von Mises, top
QERC, Haar	2.7×10^{-2}	9.8×10^{-3}	9.9×10^{-3}	2.0×10^{-2}
MLP2	2.7×10^{-2}	8.8×10^{-3}	8.1×10^{-3}	2.1×10^{-2}
tanh RF	1.5×10^{-2}	8.0×10^{-3}	3.2×10^{-3}	1.5×10^{-2}
Fourier RF	$\mathbf{6.0 \times 10^{-3}}$	$\mathbf{4.7 \times 10^{-3}}$	$\mathbf{1.5 \times 10^{-3}}$	$\mathbf{6.5 \times 10^{-3}}$

6.2 Periodic TSV array

For the quarter unit cell (Fig. 4), cosine features are best for narrow pitches ($L = 3, 4$: 1.4×10^{-5} and 2.7×10^{-5}, close to the FE reference accuracy). For wider pitches the cosine features degrade fastest ($L = 10$: 4.4×10^{-3} vs 1.0×10^{-3} for QERC). This is not a collocation-density effect (doubling the points changes nothing) but a feature-count effect: with 1024 features at $L = 6$, all models drop to $\sim 10^{-5}$ and the cosine features are again best (5.9×10^{-6}). Across all pitches, QERC and MLP2 agree within the seed spread.

6.3 Axisymmetric and 3D unit cells: the near-surface region

The region that matters for device placement is just below the wafer surface, which a 2D plane-strain model cannot represent. In the axisymmetric cell ($N = 8$, 5 seeds; Table 3) the near-surface von Mises error is 3–15 times the mid-plane error for every model, and the cosine features are the most accurate in both the clamped and the free-top case; QERC is on par with the random two-layer network. The 3D cell with the Abaqus reference shows the same ordering at $N = 10$ (clamped: QERC 2.9×10^{-2} vs Fourier 1.3×10^{-2} in $\boldsymbol{u}$; free top: 8.0×10^{-3} vs 4.4×10^{-3}).

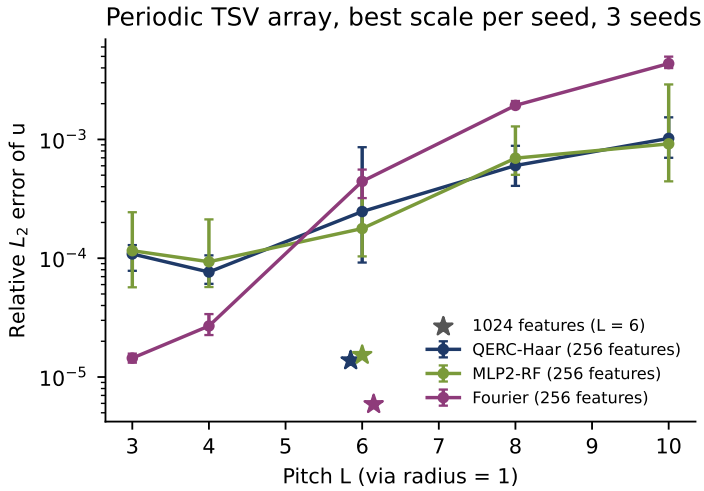


Figure 4: Periodic TSV array: displacement error versus pitch L (256 features; stars: 1024 features at $L = 6$).

Table 4: Kirsch, $N = 10$, Haar reservoir, 2 seeds: relative $L_2(\boldsymbol{u})$ versus shots per circuit. Ridge and boundary penalty chosen by the test error for all methods (optimistic bound).

Shots	Strong form	Strong, split shots	Energy (Ritz)	Energy, split shots
10^3	0.46	0.72	**0.32**	0.66
10^5	0.25	0.25	**0.083**	0.16
10^7	0.062	0.13	**0.014**	0.035
∞	6.3×10^{-4}	–	$\mathbf{2.1 \times 10^{-4}}$	–

6.4 A high-dimensional counterexample

One may expect the exponential tensor-product space to help when the input is high-dimensional. For the dispersion relation of a d-layer laminate ($d = N$ inputs), whose exact solution lies in the 2^N-dimensional span of $\bigotimes_i(\cos\varphi_i, \sin\varphi_i)$, all random feature models fail (relative error > 1), quantum or classical: a random reservoir picks a random 2^N-dimensional subspace of a 3^N–4^N-dimensional space, whose overlap with the target shrinks exponentially. A benefit would require a reservoir designed for the problem.

7 Shot noise and an energy formulation

On hardware, p_k and its derivatives are estimated from S shots per circuit, with derivatives from the parameter-shift rule [9, 10]. We propagate the multinomial variance $p(1-p)/S$ through the chain rule (an approximate model that ignores correlations between circuits) and add the noise only when assembling the system. Since $p_k \approx 2^{-N}$, the relative noise per feature scales as $\sqrt{2^N/S}$, and second derivatives amplify it.

The energy (Ritz) formulation [11] minimizes $\frac{1}{2}\int \boldsymbol{\varepsilon} : \mathbb{C} : \boldsymbol{\varepsilon}\, dA$ plus a boundary penalty, so the stiffness matrix needs only first derivatives; the traction-free hole is a natural boundary condition. It reduces the error 3–4 times at every shot level and recovers $K_t = 3.0$ from 10^5 shots, where the strong form gives 1.4 (Table 4, Fig. 5). Splitting the shots into two independent halves removes the noise-squared bias of the normal matrix but makes the results worse: the bias acts as a ridge regularizer. Fewer qubits reduce the relative noise but lose expressivity ($N = 6$ saturates at 0.2). Even so, 10^7 shots per circuit still give 1.4% error, 70 times the noiseless value.

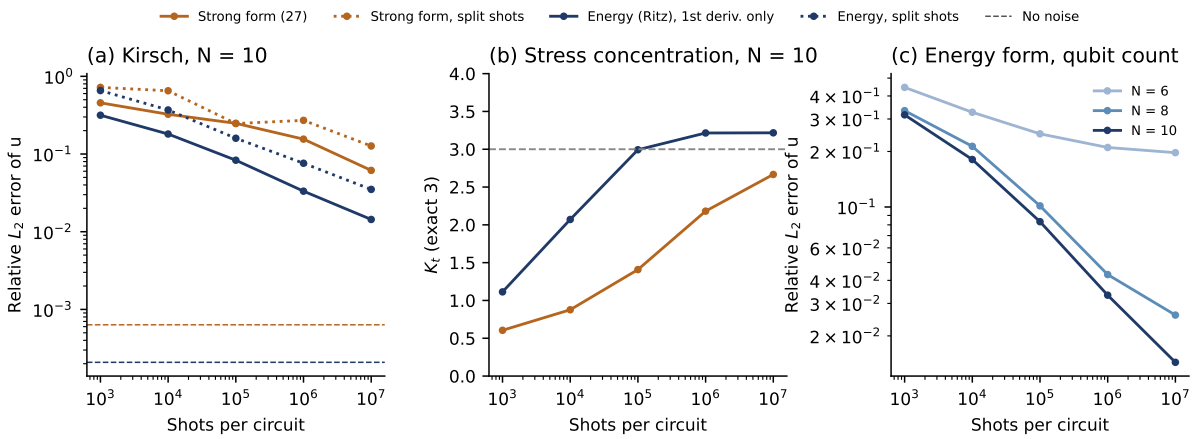


Figure 5: Shot noise: strong versus energy formulation, with and without split-shot estimation, and qubit count.

8 Discussion

- The design freedom of a quantum reservoir PINN lies in the encoding. The reservoir only needs to be generic.
- Whatever advantage appears over simple classical features is reproduced by a classical model with the same tensor-product structure; the quantum claim reduces to accessing $\mathcal{F}_\rho$ with N qubits, which matters only when 4^N classical cost is prohibitive.
- Tuning of the classical baseline changes conclusions by more than an order of magnitude and must be reported.
- Shot noise, not expressivity, limits hardware use. Formulations with low derivative order and readouts with larger probabilities (few observables instead of 2^N outcomes) are the next steps.

9 Conclusion

For linear solid mechanics, a quantum reservoir with least-squares readout is a working PINN solver, but under a fair comparison it matches classical random features with the same structure and trails cosine features. We hope the benchmark and the controls serve as a baseline for future claims.

Code and data. `https://github.com/mmc-research-group/nishioka-qpinn-solid`

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